ATF Field Specification v1.4

OMNIX QUANTUM LTD · Harold Alberto Nunes Rodelo · Day 1 Deliverable for VeriSigil AI · 22 May 2026

ATF FIELD SPECIFICATION Agent Trust Fabric — Trace Format & Field Definitions Version: 1.4 · Date: 2026-05-22 Issued by: Harold Alberto Nunes Rodelo, ATF / OMNIX QUANTUM LTD For: VeriSigil AI (Raheem Larry Babatunde) — Bridge Validation Collaboration Status: Day 1 Deliverable — Sandbox / Evaluation Only

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Overview

This document defines the complete field specification for the three core ATF record types used in the TAR↔TAP and RCR↔Survivability bridge validation:

- DR — Delegation Receipt (ATFDR-{16HEX})
- TAR — Temporal Admissibility Record (ATFTAR-{16HEX})
- RCR — Runtime Continuity Record (ATFRCR-{16HEX})

All records are PQC-signed (Dilithium-3 / ML-DSA-65, FIPS 204). All content_hash values are SHA-256 hex digests (no prefix). All timestamps are ISO 8601 UTC unless noted as nanosecond epoch.

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1. Delegation Receipt (DR)

Record ID format: `ATFDR-{16 uppercase hex characters}` ADR reference: ADR-156

1.1 Fields

Field	Type	Required	Description
`delegation_id`	string	REQUIRED	Unique DR identifier. Format: `ATFDR-{16HEX}`
`delegator_id`	string	REQUIRED	AID or human operator ID issuing the delegation
`delegate_id`	string	REQUIRED	AID of the receiving (delegated) agent
`task_scope`	object	REQUIRED	Dict describing what the delegate is authorized to do. Arbitrary key-value pairs.
`authority_budget_delegator`	float	REQUIRED	Authority budget held by delegator at issuance. Range: 0.0–100.0
`authority_budget_granted`	float	REQUIRED	Authority budget granted to delegate. MUST be ≤ `authority_budget_delegator` (ATF-INV-001 / MAR).
`parent_delegation_id`	string	OPTIONAL	DR ID of the delegation that gave delegator its authority. Null if delegator is human root.

Field	Type	Required	Description
`chain_root_id`	string	REQUIRED	DR ID of the originating human-root delegation in the chain.
`delegation_depth`	integer	REQUIRED	Chain depth. 0 = human root, 1 = first agent, N = Nth sub-agent.
`delegator_public_key`	string	REQUIRED	Dilithium-3 public key of delegator, base64-encoded. Used for offline DR signature verification. May be empty string `""` when signing key was not available at issuance time — treat empty string as unsigned.
`content_hash`	string	REQUIRED	SHA-256 hex digest of all DR fields except `content_hash`, `pqc_signature`, `pqc_algorithm`. Canonical JSON: `json.dumps(fields, sort_keys=True, separators=(",", ":"), default=str)`. Note: `authority_budget_delegator` and `authority_budget_granted` are rounded to 4dp before hashing.
`posture_state_hash`	string	CONDITIONAL	SHA-256 of authority posture at exact moment of DR issuance. Present in all DRs issued after ATF v1.0. Null only in legacy pre-v1.0 records. Committed fields: `delegator_id`, `task_scope`, `authority_budget_delegator` (rounded to 4dp), `authority_budget_granted` (rounded to 4dp), `delegation_depth`, `chain_root_id`, `created_at`. Canonical JSON: `json.dumps(fields, sort_keys=True, separators=(",", ":"))`. Enables SPV binding for TAR↔TAP bridge validation.
`pqc_signature`	string	CONDITIONAL	Dilithium-3 (ML-DSA-65) signature over `content_hash`, base64-encoded. Present when signing key is available.
`pqc_algorithm`	string	CONDITIONAL	Signing algorithm identifier. Value: `"dilithium3"`
`expires_at`	string	OPTIONAL	ISO 8601 UTC expiry timestamp. Null = no expiry.
`status`	string	REQUIRED	`ACTIVE | EXPIRED | REVOKED`
`created_at`	string	REQUIRED	ISO 8601 UTC timestamp of DR issuance.
`metadata`	object	REQUIRED	Extension dict. Empty object `{}` if unused.

1.2 Invariants

- ATF-INV-001 (MAR): `authority_budget_granted` ≤ `authority_budget_delegator`. Violation raises hard error before issuance — no DR is created.

- ATF-INV-002: `chain_root_id` traces back to a human-root delegation (`delegation_depth = 0`).
- ATF-INV-003: `posture_state_hash` is computed before `content_hash`. `content_hash` commits to `posture_state_hash`.

1.3 Example

```
{
  "delegation_id": "ATFDR-A1B2C3D4E5F60001",
  "delegator_id": "AID-COMPLIANCE-A1B2C3D4E5F60001",
  "delegate_id": "AID-AUDIT-F1E2D3C4B5A60007",
  "task_scope": {
    "domain": "compliance",
    "permitted_actions": ["read_evidence", "issue_tar"],
    "max_duration_s": 3600
  },
  "authority_budget_delegator": 80.0,
  "authority_budget_granted": 60.0,
  "parent_delegation_id": "ATFDR-ROOT0000000001",
  "chain_root_id": "ATFDR-ROOT0000000001",
  "delegation_depth": 1,
  "delegator_public_key": "<base64-dilithium3-pubkey>",
  "content_hash": "a3f1c2d4e5b6a7f8c9d0e1f2a3b4c5d6e7f8a9b0c1d2e3f4a5b6c7d8e9f0a1b2",
  "posture_state_hash": "28c1dcbd36bae0faf708af58f72d5cc5103974266b3ab47c52cf76c23bcb1095",
  "pqc_signature": "<base64-dilithium3-signature>",
  "pqc_algorithm": "dilithium3",
  "expires_at": "2026-05-22T10:00:00+00:00",
  "status": "ACTIVE",
  "created_at": "2026-05-21T10:00:00+00:00",
  "metadata": {}
}
```

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2. Temporal Admissibility Record (TAR)

Record ID format: `ATFTAR-{16 uppercase hex characters}` ADR reference: ADR-157

2.1 Fields

Field	Type	Required	Description
`tar_id`	string	REQUIRED	Unique TAR identifier. Format: `ATFTAR-{16HEX}`
`delegation_id`	string	REQUIRED	`ATFDR-{16HEX}` of the authorizing Delegation Receipt
`agent_id`	string	REQUIRED	AID of the acting agent. Format: `AID-{DOMAIN}-{16HEX}`
`execution_ref`	string	OPTIONAL	Reference to Execution Receipt (ADR-131). Null if not linked.
`execution_ns`	integer	REQUIRED	Nanosecond Unix timestamp of the execution event being admitted.
`execution_ts`	string	REQUIRED	ISO 8601 UTC timestamp of the execution event.

Field	Type	Required	Description
`dr_status_at_admission`	string	REQUIRED	Status of the DR at the moment of TAR admission. Value: `ACTIVE` | `EXPIRED` | `REVOKED`
`dr_expires_at`	string	OPTIONAL	DR expiry timestamp, copied from the DR.
`authority_budget`	float	REQUIRED	Authority budget of the DR at admission time.
`domain`	string	REQUIRED	Governance domain identifier (e.g. `"compliance"`, `"trading"`, `"medical_ai"`).
`task_action`	string	REQUIRED	The specific action being admitted (e.g. `"read_evidence"`, `"issue_receipt"`).
`admission_status`	string	REQUIRED	`ADMITTED | REJECTED`
`rejection_reason`	string	OPTIONAL	Human-readable reason for rejection. Null if admitted.
`content_hash`	string	REQUIRED	SHA-256 hex digest of all TAR fields except `content_hash`, `pqc_signature`, `pqc_algorithm`. Canonical JSON: `json.dumps(fields, sort_keys=True, separators=(",", ":"), default=str)`.
`pqc_signature`	string	CONDITIONAL	Dilithium-3 (ML-DSA-65) signature over `content_hash`, base64-encoded.
`pqc_algorithm`	string	CONDITIONAL	`"dilithium3"`
`chain_root_id`	string	REQUIRED	`chain_root_id` of the authorizing DR. Enables chain-level traceability.
`issued_at`	string	REQUIRED	ISO 8601 UTC timestamp of TAR issuance.
`metadata`	object	REQUIRED	Extension dict.

2.2 Invariants

- ATF-INV-004: A TAR with `admission_status = ADMITTED` MUST reference a DR with `dr_status_at_admission = ACTIVE`.
- ATF-INV-005: `execution_ns` MUST be ≤ the nanosecond timestamp of `issued_at`.
- TAR-INV-006: DR remaining validity at TAR admission MUST NOT exceed `TAR_MAX_DR_LIFETIME_SECONDS` (86400 s / 24 h). This is a **compiled structural constant** — not configurable by operator or environment variable. Clock skew tolerance: 5 s. DRs with > 24 h remaining at admission time are automatically `REJECTED`. To issue authority beyond 24 h, a fresh DR must be issued, creating a new auditable issuance event.
- TAR is the primary bridge target for TAR↔TAP equivalence mapping. `delegation_id` is the join key.

2.3 Example

```
{
  "tar_id": "ATFTAR-F1E2D3C4B5A60001",
  "delegation_id": "ATFDR-A1B2C3D4E5F60001",
  "agent_id": "AID-AUDIT-F1E2D3C4B5A60007",
  "execution_ref": null,
```

```
  "execution_ns": 1748131200000000000,
  "execution_ts": "2026-05-21T10:00:00+00:00",
  "dr_status_at_admission": "ACTIVE",
  "dr_expires_at": "2026-05-22T10:00:00+00:00",
  "authority_budget": 60.0,
  "domain": "compliance",
  "task_action": "read_evidence",
  "admission_status": "ADMITTED",
  "rejection_reason": null,
  "content_hash": "c5d6e7f8a9b0c1d2e3f4a5b6c7d8e9f0a1b2c3d4e5f6a7b8c9d0e1f2a3b4c5d6",
  "pqc_signature": "<base64-dilithium3-signature>",
  "pqc_algorithm": "dilithium3",
  "chain_root_id": "ATFDR-ROOT0000000001",
  "issued_at": "2026-05-21T10:00:01+00:00",
  "metadata": {}
}
```

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3. Runtime Continuity Record (RCR)

Record ID format: `ATFRCR-{16 uppercase hex characters}` ADR reference: ADR-159

3.1 Fields

Field	Type	Required	Description
`rcr_id`	string	REQUIRED	Unique RCR identifier. Format: `ATFRCR-{16HEX}`
`tar_id`	string	REQUIRED	`ATFTAR-{16HEX}` of the admission record this RCR monitors.
`delegation_id`	string	REQUIRED	`ATFDR-{16HEX}` of the authorizing DR.
`chain_root_id`	string	REQUIRED	Chain root delegation ID.
`agent_id`	string	REQUIRED	AID of the running agent.
`ces_score`	float	REQUIRED	Continuity Eligibility Score. Range: 0.0–100.0. Formula: `(T×0.30) + (B×0.30) + (D×0.20) + (I×0.20)`. Rounded to 4dp.
`continuity_status`	string	REQUIRED	`NOMINAL` (75–100) | `MONITORING` (50–75) | `WARNING` (25–50) | `CRITICAL` (10–25) | `HALT` (0–10)
`ces_temporal`	float	REQUIRED	T component: `(time remaining on DR / total DR lifetime) × 100`. Range: 0.0–100.0. Defaults to 100.0 when DR has no expiry (`dr_expires_at` is null). Rounded to 4dp.
`ces_budget`	float	REQUIRED	B component: `(budget remaining / budget at admission) × 100`. Range: 0.0–100.0. Defaults to 100.0 when `budget_at_admission = 0`. Rounded to 4dp.
`ces_context`	float	REQUIRED	D component: `100 - context_drift_pct`. Range: 0.0–100.0. Rounded to 4dp.
`ces_integrity`	float	REQUIRED	I component: `100 - (active_anomalies × 10)`. Range: 0.0–100.0. Rounded to 4dp.

Field	Type	Required	Description
`time_remaining_ns`	integer	OPTIONAL	Remaining DR lifetime in nanoseconds at this snapshot. Null when DR has no expiry (`dr_expires_at` is null).
`dr_expires_at`	string	OPTIONAL	ISO 8601 UTC expiry timestamp of the DR. Null if DR has no expiry.
`fragmentation_score`	float	REQUIRED	Aggregate budget consumption % across all active sessions sharing the same `chain_root_id` . Formula: `(total_admitted - total_remaining) / total_admitted × 100` . Range: 0.0–100.0
`reauth_challenge_id`	string	OPTIONAL	Reference to ReauthorizationChallenge if issued (`ATFRC-{16HEX}`). Null otherwise.
`execution_ns`	integer	REQUIRED	Nanosecond Unix timestamp of the execution event.
`execution_ts`	string	REQUIRED	ISO 8601 UTC timestamp of the execution event.
`issued_at`	string	REQUIRED	ISO 8601 UTC timestamp of RCR issuance.
`budget_at_admission`	float	REQUIRED	Authority budget at the moment of TAR admission.
`budget_remaining`	float	REQUIRED	Authority budget remaining at this RCR snapshot.
`context_drift_pct`	float	REQUIRED	Context drift percentage at this snapshot. Range: 0.0–100.0
`active_anomalies`	integer	REQUIRED	Count of active governance anomalies at this snapshot.
`content_hash`	string	REQUIRED	SHA-256 hex digest of all RCR fields except `content_hash` , `pqc_signature` , `pqc_algorithm` . Canonical JSON: `json.dumps(fields, sort_keys=True, separators=(",", ":"), default=str)` .
`pqc_signature`	string	CONDITIONAL	Dilithium-3 (ML-DSA-65) signature over `content_hash` , base64-encoded.
`pqc_algorithm`	string	CONDITIONAL	`"dilithium3"`
`sample_reason`	string	OPTIONAL	Reason for this RCR snapshot: `SCHEDULED` | `EXECUTION_COMPLETE` | `THRESHOLD_BREACH` | `FRAGMENTATION` | `EXTERNAL` | `RC_TTL_EXPIRED` .
`escalation_event_id`	string	OPTIONAL	ID of ContinuityEscalationEvent if triggered. Format: `ATFCEE-{16HEX}` . Null otherwise.
`predecessor_rcr_id`	string	OPTIONAL	ID of previous RCR in the continuity chain.
`metadata`	object	REQUIRED	Extension dict.

3.2 Invariants

- RGC-INV-001: Every RCR MUST be anchored to a valid TAR (`tar_id` must resolve).
- RGC-INV-002: CES MUST be computed from real-time values only — no cached inputs.

- RGC-INV-003: `HALT` tier terminates execution and revokes in-flight sub-tasks.
- RGC-INV-005: All RCRs are PQC-signed and immutable after issuance.
- RGC-INV-006: RCR chain is acyclic — `issued_at` MUST be strictly monotonically increasing per execution session.
- RCR is the bridge target for RCR↔Survivability receipt mapping. Join key: `tar_id`.

3.3 Example


```
{
  "rcr_id": "ATFRCR-D1C2B3A4E5F60001",
  "tar_id": "ATFTAR-F1E2D3C4B5A60001",
  "delegation_id": "ATFDR-A1B2C3D4E5F60001",
  "chain_root_id": "ATFDR-ROOT0000000001",
  "agent_id": "AID-AUDIT-F1E2D3C4B5A60007",
  "ces_score": 92.7,
  "continuity_status": "NOMINAL",
  "ces_temporal": 87.0,
  "ces_budget": 92.0,
  "ces_context": 95.0,
  "ces_integrity": 100.0,
  "execution_ns": 1748131200000000000,
  "execution_ts": "2026-05-21T10:00:00+00:00",
  "issued_at": "2026-05-21T13:07:12+00:00",
  "budget_at_admission": 60.0,
  "budget_remaining": 55.2,
  "context_drift_pct": 5.0,
  "active_anomalies": 0,
  "time_remaining_ns": 75168000000000,
  "dr_expires_at": "2026-05-22T10:00:00+00:00",
  "fragmentation_score": 8.0,
  "reauth_challenge_id": null,
  "content_hash": "d6e7f8a9b0c1d2e3f4a5b6c7d8e9f0a1b2c3d4e5f6a7b8c9d0e1f2a3b4c5d6e7",
  "pqc_signature": "<base64-dilithium3-signature>",
  "pqc_algorithm": "dilithium3",
  "sample_reason": "SCHEDULED",
  "escalation_event_id": null,
  "predecessor_rcr_id": null,
  "metadata": {}
}
```


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4. Cross-Record Traceability

The three records form a traceable chain:

```
DR (ATFDR-...)
  └── TAR (ATFTAR-...) — references delegation_id
        └── RCR (ATFRCR-...) — references tar_id + delegation_id
```

Join path for TAR↔TAP bridge: `TAR.delegation_id` → `DR.delegation_id` → `DR.posture_state_hash`

Join path for RCR↔Survivability bridge: `RCR.tar_id` → `TAR.tar_id` → `TAR.delegation_id` → `DR.chain_root_id`

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5. Signature Verification

Steps 1–3 are identical for all three record types:

1. Recompute `content_hash` from all fields except `content_hash`, `pqc_signature`, `pqc_algorithm`
2. Canonical JSON: `json.dumps(fields, sort_keys=True, separators=(",", ":"), default=str)`
3. SHA-256 hex of UTF-8 encoded canonical JSON

Step 4 differs by record type:

DR: Verify Dilithium-3 signature over `content_hash.encode("utf-8")` using `DR.delegator_public_key` (the key is embedded in the DR itself — no external key required).

TAR and RCR: Verify Dilithium-3 signature over `content_hash.encode("utf-8")` using the OMNIX platform public key.

OMNIX platform public key (ML-DSA-65 / Dilithium-3, base64): Shared separately. See integration comms for current value.

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6. Domain Values (Common)

Recognized `domain` values:

Value	Description
`compliance`	ISO 27001 / regulatory compliance
`trading`	Autonomous trading governance
`medical_ai`	Medical AI recommendation governance
`defense_governance`	Defense / national security AI
`insurance`	Insurance underwriting AI
`real_estate`	Real estate AI decisions
`energy_governance`	Energy sector AI
`islamic_credit`	Islamic finance credit AI
`stablecoin`	Stablecoin / DeFi governance
`crisis`	Crisis response AI
`autonomous_agent`	General autonomous agent
`robotics`	Robotics governance

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7. Revision History

Version	Date	Change
1.0	2026-05-21	Initial release — Day 1 deliverable for VeriSigil bridge validation
1.1	2026-05-22	

Version	Date	Change
		RCR field names corrected to match implementation: ces→ces_score, ces_tier→continuity_status, temporal_health→ces_temporal, budget_health→ces_budget, context_fidelity→ces_context, integrity_score→ces_integrity. Added budget_at_admission, budget_remaining, context_drift_pct, active_anomalies, sample_reason, escalation_event_id, predecessor_rcr_id.
1.2	2026-05-22	CES component ranges corrected: 0.0–1.0 → 0.0–100.0 for ces_temporal, ces_budget, ces_context, ces_integrity. Component formulas corrected (no / 100 division). Added missing RCR fields: time_remaining_ns, dr_expires_at, fragmentation_score, reauth_challenge_id. Example JSON values corrected. Fixed RGC-INV-006 reference from issued_at_ns to issued_at. DR signature verification corrected: uses DR.delegator_public_key, not platform key. TAR/RCR use platform key. Canonical JSON updated to include default=str for all three record types.
1.3	2026-05-22	sample_reason values corrected: removed ANOMALY_DETECTED/MANUAL, added THRESHOLD_BREACH/FRAGMENTATION/EXTERNAL/RC_TTL_EXPIRED. fragmentation_score range corrected: 0.0–1.0 → 0.0–100.0, formula documented. Example fragmentation_score corrected: 0.08 → 8.0. posture_state_hash: added float rounding (4dp), canonical JSON format, status changed to CONDITIONAL. ces_score example corrected: 82.5 → 92.7. posture_state_hash example hash replaced with real computed value. DR signature uses delegator_public_key (not platform key); added empty-string note. CES components: documented 4dp rounding and ces_temporal=100.0 default when no expiry. time_remaining_ns corrected to OPTIONAL. escalation_event_id format ATFCEE-{16HEX} added. robotics domain added. canonical JSON updated to include default=str for content_hash in all three record types.
1.4	2026-05-22	TAR-INV-006 added: compiled 24h DR lifetime staleness bound (TAR_MAX_DR_LIFETIME_SECONDS=86400, not configurable). DR example expires_at corrected: 26h → 24h (was violating TAR-INV-006 and would auto-REJECT). All three examples updated to 24h expiry. RCR example time_remaining_ns corrected: 93600000000000 → 75168000000000 (consistent with ces_temporal=87.0 over 24h DR). RCR example issued_at corrected: 10:01:00 → 13:07:12 (consistent with time_remaining_ns). DR example delegator_id corrected to 16-hex format: AID-COMPLIANCE-A1B2C3D4E5F60001. DR example delegate_id corrected to 16-hex format: AID-AUDIT-F1E2D3C4B5A60007. TAR and RCR agent_id aligned with DR delegate_id. posture_state_hash recomputed with corrected delegator_id. TAR and RCR content_hash descriptions updated to include default=str. ces_score documented as rounded to 4dp.